

Data Set Source:

https://www.basketball-reference.com/leagues/NBA_2026_advanced.html

Research Question:

Which individual player statistics best predict their team's performance when they're on the court (as measured by plus/minus)?

Target Variable: Box Plus/Minus

Predictors: Position 1, Age, Points per game, Assists, Rebounds

Response variable: Box plus minus

Introduction:

Understanding which individual player statistics best predict their team's performance when they're on the court is a key question in modern basketball analytics. Traditional scoring metrics often fail to fully capture a player's overall impact on winning, which has led to the development of more comprehensive measures such as Box Plus/Minus (BPM). BPM estimates a player's contribution to a team's scoring margin per 100 possessions by incorporating box-score statistics like points, assists, rebounds, and defensive actions. Because it combines both offensive and defensive contributions into a single value, BPM serves as a clear and practical response variable for evaluating player effectiveness.

In this project, we consider several predictors, including player position, age, points per game, assists, rebounds, and steals. These variables are widely recognized as indicators of offensive output, playmaking ability, and defensive performance. Ghimire, Ehrlich, and Sanders (2020) applied regularized regression methods to Real Plus-Minus data and found that both player characteristics and lineup context significantly explain variation in plus-minus outcomes. Their findings highlight the usefulness of multivariable approaches in isolating individual contributions within a team setting.

Similarly, Gong and Su (2024) introduced hierarchical models to estimate positional plus-minus in the NBA, demonstrating that both player position and performance metrics systematically

influence plus-minus ratings. Their work reinforces the importance of including positional and statistical predictors when analyzing player impact. Additionally, Sarlis (2024) employed data mining and multivariate techniques to examine NBA performance indicators and concluded that traditional box-score statistics remain strong predictors of advanced metrics like BPM.

Together, these studies suggest a strong relationship between observable player statistics and plus-minus measures. Linear regression is well suited for analyzing these relationships, as it allows for the simultaneous evaluation of multiple predictors, produces interpretable coefficients that quantify effect sizes, and enables hypothesis testing for each variable. The primary goal of this project is to determine which performance measures most strongly influence BPM, with implications for player evaluation, team strategy, and talent development in professional basketball.

Data Description:

We used an open-source NBA player statistics from Basketball-Reference for the 2024-2025 and 2025–26 regular season. Player box-score variables were taken from the “NBA_2026 Per Game” table, and BPM was taken from the “NBA_2026 Advanced” table. These tables were merged at the player-season level so each observation represents one player’s performance in either the 2024-25 or 2025-26 season.

Sports Reference compiles basketball statistics from various sources, contributors, and collaborators and maintains documented coverage across seasons. Originally, the purpose of the dataset was to be a historical meaning, a statistical summary of a player's tenure in the NBA, while we repurposed the dataset for inference: identifying which common box-score measures are most strongly associated with a player’s impact metric.

Our response variable is BPM which is defined as points-per-100-possession scale, so positive values indicate above-average impact. BPM has a mean of -1.26 (SD: 3.63), ranging from -16.2 to 15.8, reflecting wide variation in player impact across the league. Additionally, we came to the conclusion that the usage of multi-linear regression, as the residuals versus fitted plot shows no strong systematic curvature, suggesting the mean relationship is approximately linear. However, the scale–location plot shows a noticeable pattern consistent with non-constant

variance, so classical constant-variance assumptions may not hold exactly. The Q–Q plot indicates residuals are close to normal for most observations but deviate in the tails, implying inference could be mildly sensitive to extreme players. The residuals-versus-leverage plot does not indicate that a large number of points are highly influential, though any high-leverage cases should be checked before final conclusions.

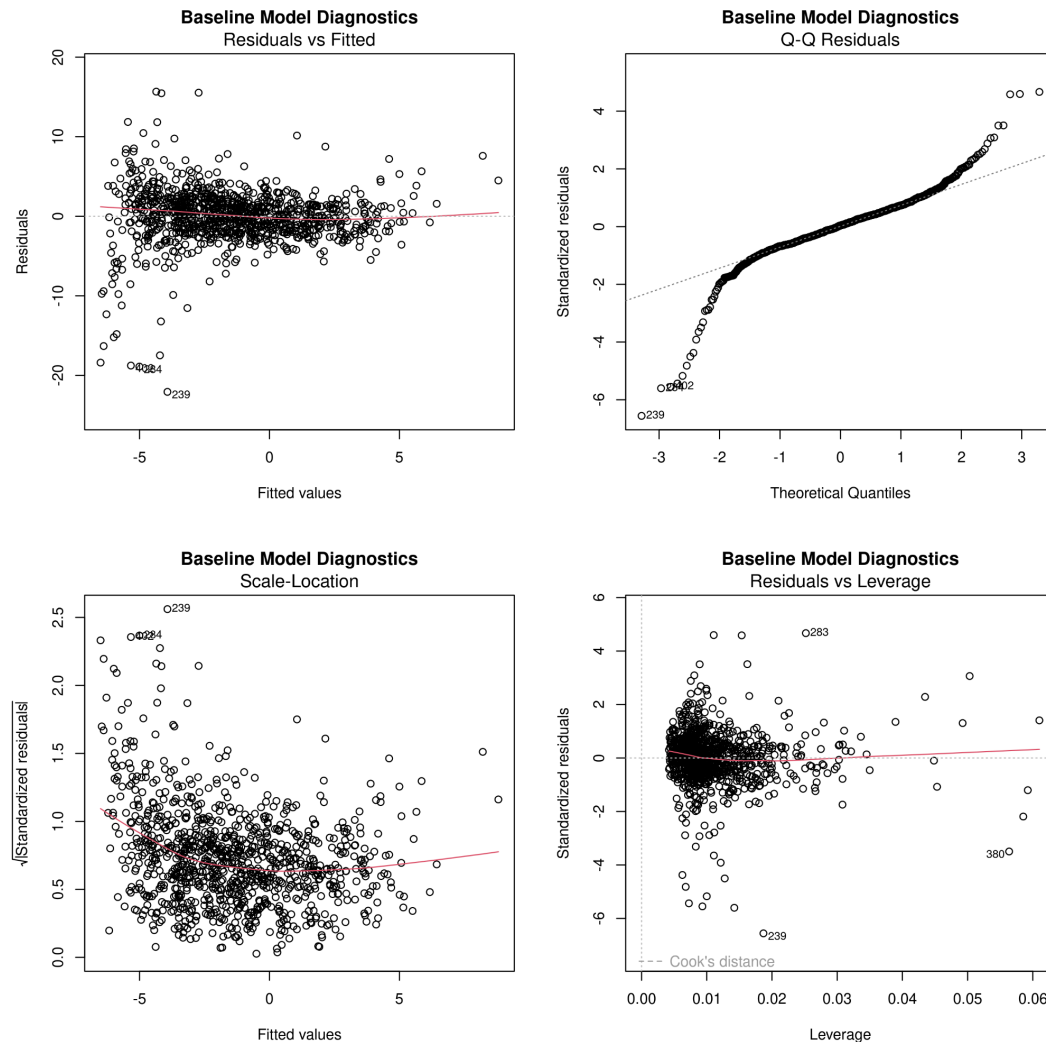
Lastly, predictors in the preliminary model are Age, per-game PTS, AST, ORB, BLK and STL. Moreover, age varies widely (mean 25.99, SD 4.31; 19 – 41), capturing both young and veteran roles. PTS (9.09; 0 – 32.8), and AST (2.15; 0 – 10.7) have long right tails with many near-zero players, suggesting outliers. ORB (0.95; 0 – 4.7) shows moderate spread, reflecting specialist contributions to BPM. STL (0.69; 0 – 3.0) and BLK (0.40; 0 – 2.7) both exhibit right-skewed distributions with many near-zero values, consistent with defensive actions being concentrated among a subset of players. Summary statistics for all predictors are displayed in the table below.

Predictor	Mean	SD	Min	Max	Description
Age	25.99	4.31	19	41	Experience and maturity
PTS	9.09	6.68	0	32.8	Scoring output per game
ORB	0.95	0.79	0	4.7	Second-chance creation
AST	2.15	1.85	0	10.7	Playmaking ability
STL	0.69	0.43	0	3	Defensive activity
BLK	0.4	0.39	0	2.7	Shot prevention

Preliminary Result:

The residuals versus fitted values plot showcases a slight funnel pattern with an approximately horizontal trendline across the entire range, suggesting that the linearity assumption is valid and the model is linear in its parameters, which is consistent with the findings of Ghimire et al. (2020) and Gong et al. (2024).

When examining the Q-Q plot for the data, there appears to be some deviations from the reference line near the tails. However, the proportion of these tails compared to the rest of the data set is relatively small and the majority of the data appears to follow the reference line, providing strong supporting evidence for the normality of data which is also noted by Sarlis and their peers (2024).



The primary model includes seven predictors, these being Age, Position, PTS, ORB, AST, STL, and BLK. Steals has the largest coefficient (1.865), meaning every steal per game corresponds to an expected increase of 1.865 BPM points, holding all other predictors constant. This points to defensive playmaking and turnover generation having a strong connection with team performance among the statistics considered.

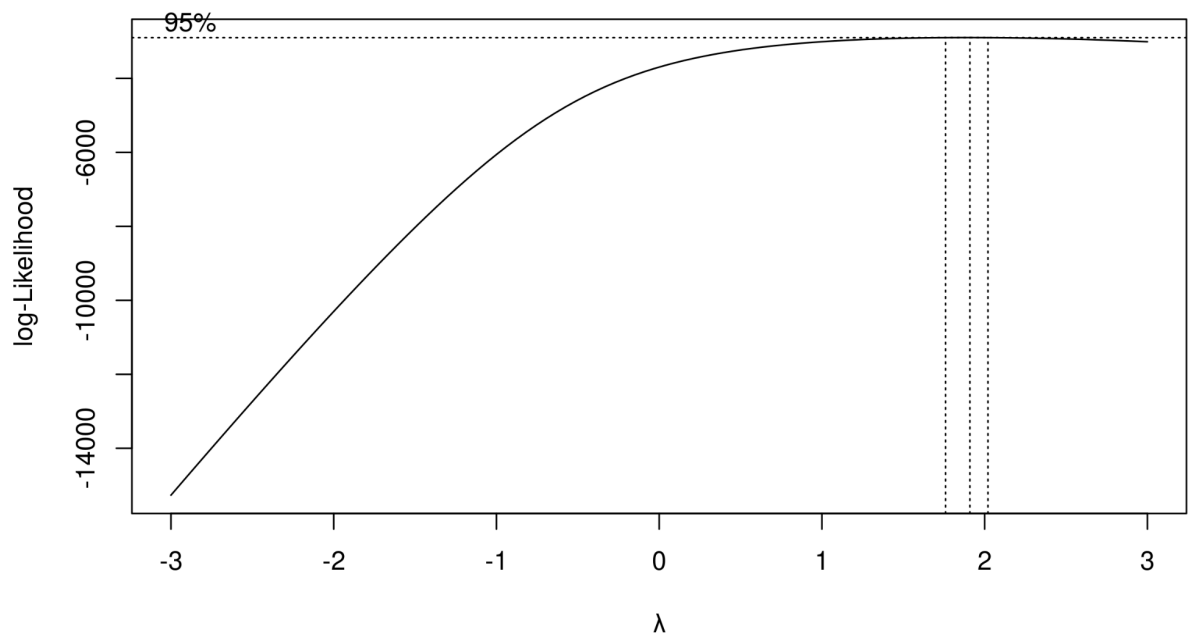
Blocks (0.834) suggests that shot prevention matters, showing that players who alter shots at the rim tend to have outsized impact. Points per game has a positive coefficient of 0.169, meaning each additional point scored per game corresponds to a BPM increase of 0.169, suggesting scoring contributes positively but to a lesser degree than defensive actions. Assists (0.238) and offensive rebounds (0.345) are both positively associated with BPM, consistent with the importance of playmaking and second-chance opportunities respectively. Age has a small positive effect (0.091), suggesting veteran players tend to have marginally higher BPM holding statistics constant.

The reference category for position was set as Centre, which had the highest BPM relative to other roles. The role type with the largest drop in BPM compared to Centres was Point Guards (-1.246), followed by Small Forwards (-1.100) and Shooting Guards (-1.071). Power Forwards showed the least drop off compared to Centres (-0.532), indicating the greatest similarities between Power Forwards and Centres.

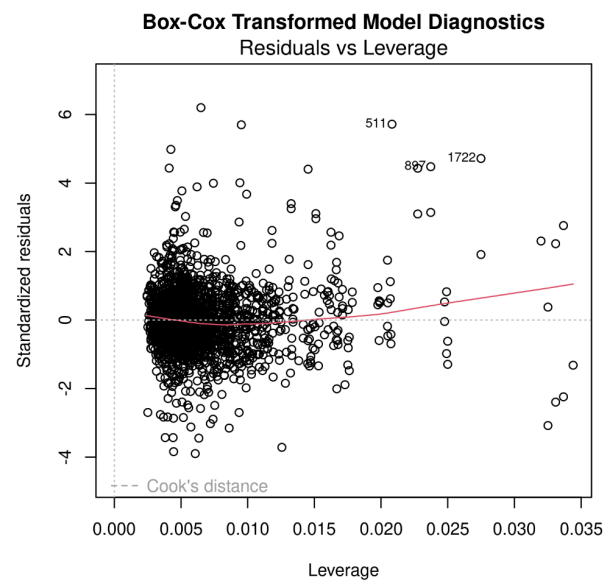
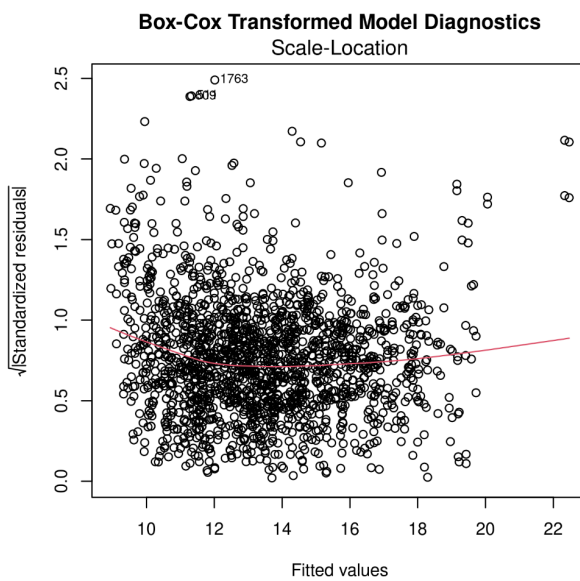
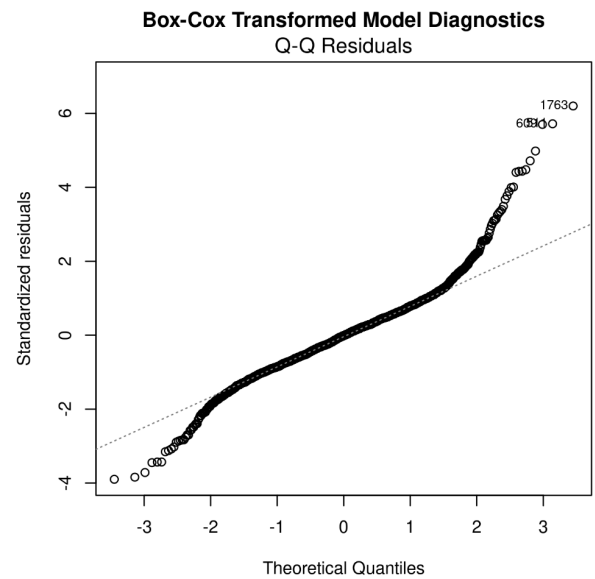
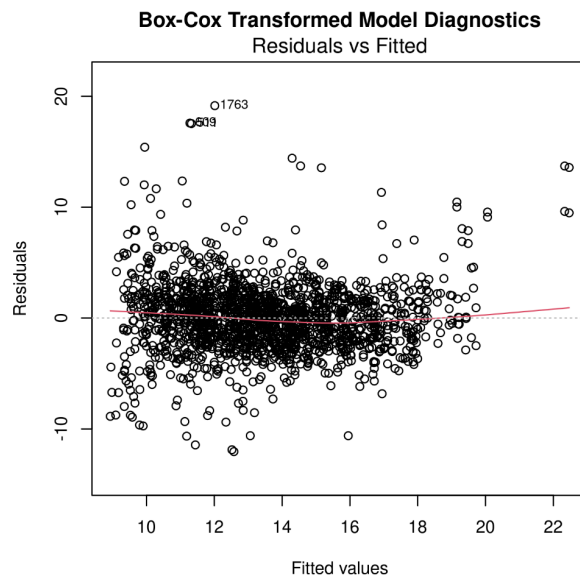
Model Selection

To determine whether a transformation of either the response or predictor variables was needed, a closer look was taken at the model's diagnostic plots. As noted in Section 3, the baseline model diagnostics revealed a funnel pattern in the Residuals vs. Fitted plot and tail departures in the Q-Q plot. The Scale-Location smoother slopes upward where it should be flat, and the Q-Q tails deviate substantially at both ends. Together, these suggest that the non-constant variance and non-normality assumptions were violated. This motivated an investigation into transforming the response variable, BPM, specifically by using the Box-Cox method.

Since the BPM takes negative values, a constant shift of $|\min(\text{BPM})| + 1$ was applied to ensure that the response variable stayed strictly positive before utilizing the method. The results of this are displayed below:



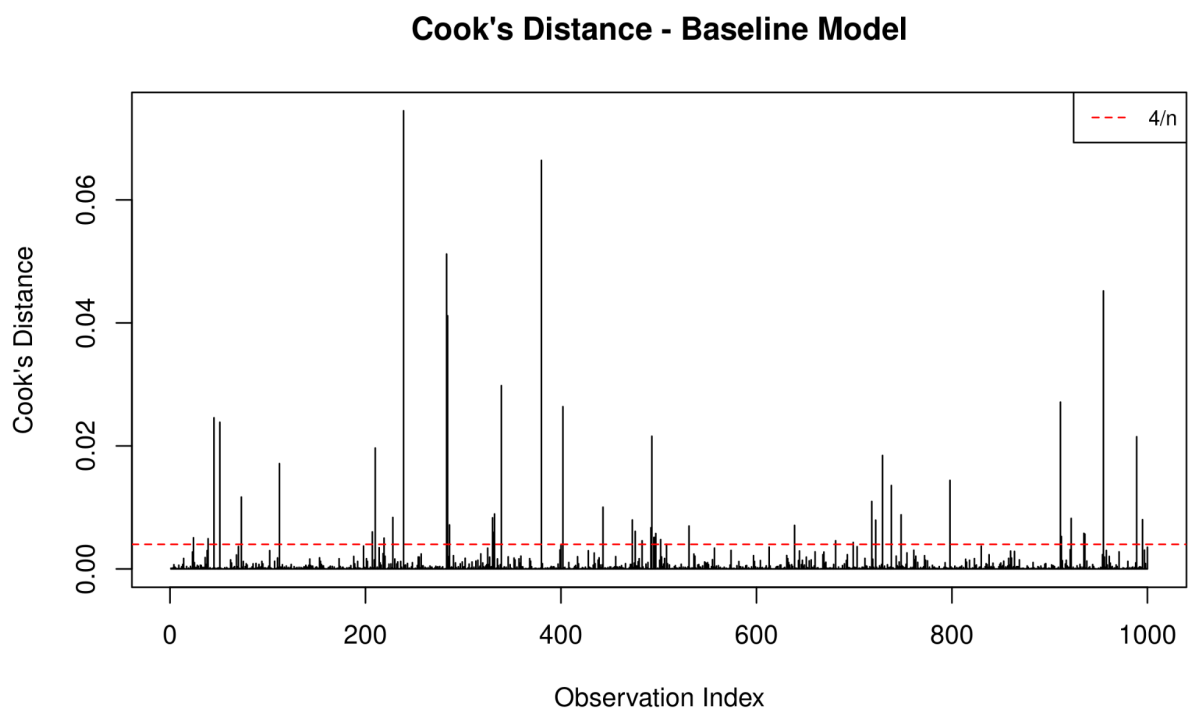
The log-likelihood profile peaks at $\lambda = 1.909$, with the 95% confidence interval falling entirely above $\lambda = 1$. This means that the identity transformation would be insufficient as a transformation. Since $\lambda = 2$ falls within the confidence interval, λ was rounded to 2, which corresponds to a square transformation. This meant that the response variable would need to be squared to, in theory, satisfy the normality and homoscedasticity assumptions according to the Box-Cox criterion. However, evaluating the Box-Cox model and examining the diagnostic plots told us a rather different story.



The Residuals vs Fitted plot still retains a funneling pattern that narrows from left to right, indicating that the constant variance assumption is still not met. This is further supported by the Scale-Location plot's smoother remaining upward sloping. Furthermore, the tails of the Q-Q diagnostic plot remain heavy, suggesting that the normality assumption violation also persists with similar severity to the baseline. Additionally, by enacting the transformation and squaring the BPM, coefficients are rendered uninterpretable in basketball terms as changes are now described in a shifted, rescaled, squared quantity rather than the original BPM units. These facts combined solidified our decision to retain the untransformed model.

The residual violations are attributed to structural heterogeneity in the data: a small number of elite players produce statistical profiles far removed from the bulk of the distribution, creating heavy tails and non-constant variance that are properties of the population rather than correctable modelling failures.

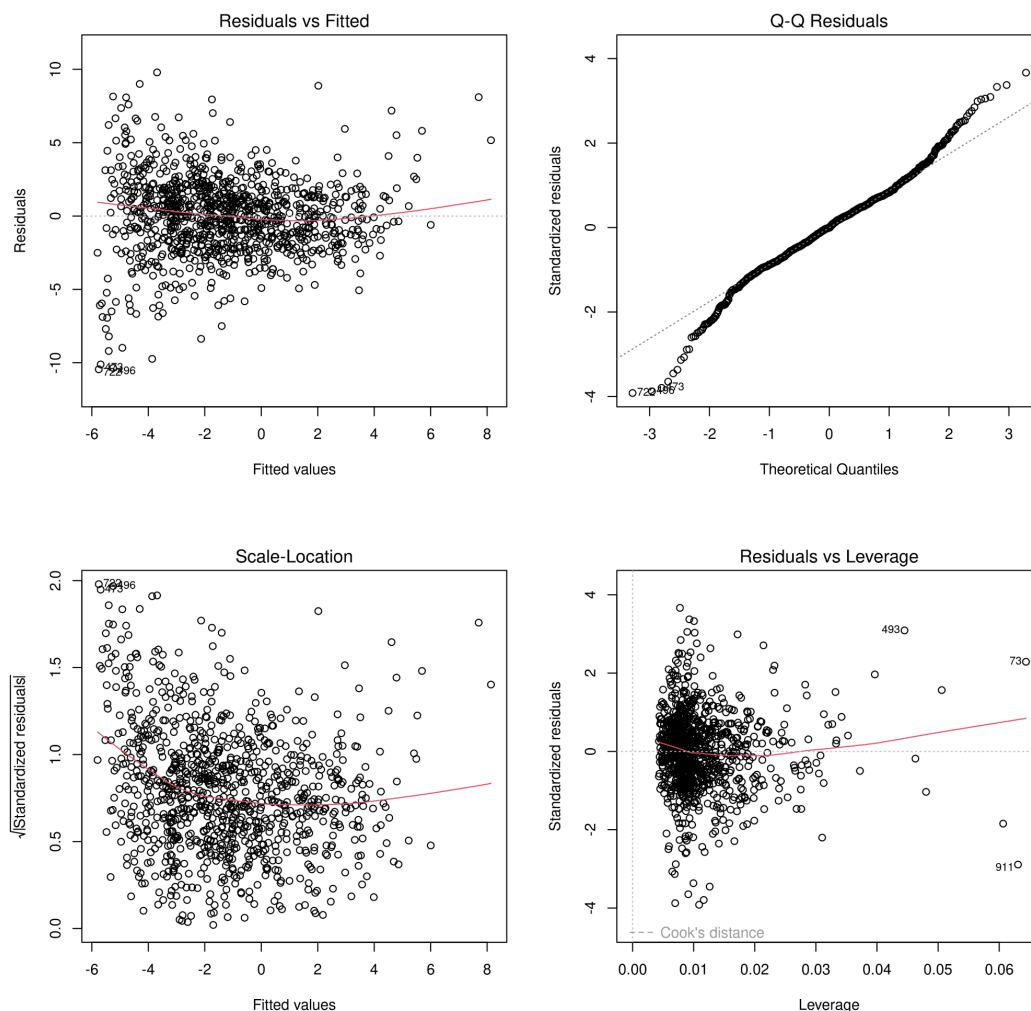
To identify and remove observations that had undue influence on the model, the Cook's Distance, studentized residuals, and leverage diagnostics were used. Cook's distance measures the aggregate shift in all fitted values upon deletion of an observation. Studentized residuals assess how far each response deviates from its fitted value, relative to the model's standard error. Leverage quantifies how unusual an observation's predictor values are relative to the rest of the data. The fact that a high leverage point can steer the regression line without even possessing a large residual prompted us to consider all three diagnostics together.



The Cook's distance plot shows many observations exceeding the conventional $4/n$ threshold, indicating that several observations are statistically unusual. These points may not necessarily be harmful to the model; however, this is why a dual criterion was applied to guard against unnecessary removal. An observation was only removed if it simultaneously exceeded

Cook's distance of $4/n$ and had an absolute studentized residual greater than 3. This ensured that only observations that were both highly influential and genuinely poorly fitted were removed.

Applying this criterion to the 1000 observations in our dataset led to the removal of 19 observations, leaving 981 player-season records. These removals are tied to players logging very few minutes of playtime, yet possessing stats that BPM would evaluate quite differently from typical contributors. Had they remained included, it would have resulted in the distortion of our coefficient estimates for the broader population of NBA players. The diagnostic plots for the final model with cleaned data support that removing these outliers was beneficial. The Residuals vs Fitted graph is smoother, flatter, and closer to zero across the range of fitted values. The Q-Q plot, while possessing some tail departure, is considerably less severe than the baseline. The plots are shown below:



The summary of the final model confirms that the vast majority of predictors included in the model are statistically significant. STL ($p < 0.001$), PTS ($p < 0.001$), Age ($p < 0.001$), Block ($p = 0.007$), AST ($p = 0.009$) and ORB ($p = 0.038$) do meet the conventional significance threshold. Furthermore, most position indicators are also significant, the sole exception being the Power Forward predictor, PosPF ($p = 0.097$). This has to be tolerated as individual levels of a categorical predictor can not be selectively dropped as it would discard meaningful information about the differences between roles as it pertains to BPM.

To verify whether one defensive predictor, such as STL needed to be excluded, a reduced model removing it was compared against the full model. AIC favoured the full model, with the full model having an AIC score of 4732 compared to 4771. The adjusted R-squared value was also higher, with a value of 0.453 compared to 0.431. This confirmed that STL contributes meaningfully to the regression model.

This was repeated for two other defensive predictors, blocks (BLK) and defensive rebounds (DRB). Blocks directly prevent scoring opportunities while defensive rebounds deny opponents a second chance to score points, both of which should be captured by BPM in theory. When BLK was added to the model, it attained statistical significance and the AIC of the model improved, validating its retention in the model. The same can not be said for DRB, which was also tested but failed to reach statistical significance and did not improve the AIC, showing that it contributed no meaningful information that other predictors did not already capture. Thus, it was excluded.

Final Model and Inferences:

Table summary of the model featuring coefficients, standard errors, confidence intervals and associated p values. The confidence intervals are calculated with 95% confidence.

Predictor	Coefficient	Std_Error	CI_Lower	CI_Upper	P_Value
(Intercept)	-6.80016272	0.667649726	-8.110366975	-5.489958467	3.24E-23
Age	0.09104726	0.020605944	0.050609895	0.131484624	1.11E-05
PosPF	-0.53171035	0.32030918	-1.160289133	0.096868427	0.09723948
PosPG	-1.24553913	0.420654197	-2.071036242	-0.420042025	0.00314153
PosSF	-1.09973866	0.356731844	-1.799793738	-0.399683588	0.00210847
PosSG	-1.07106441	0.367017255	-1.791303705	-0.350825113	0.00360079
PTS	0.16912145	0.023012073	0.12396227	0.214280634	4.23E-13
ORB	0.34486097	0.165619106	0.019847942	0.669873992	0.03758092
AST	0.23832252	0.090583031	0.060561236	0.416083805	0.00864883
STL	1.86499471	0.290327592	1.295252174	2.43473724	2.08E-10
BLK	0.83393815	0.310049354	0.225493382	1.442382917	0.00727453

The model's performance will be assessed using four metrics namely R-squared, Adjusted R-squared, Mean Absolute Error, and Root Mean Squared Error. Each provides a unique insight on the model's performance and its robustness.

R-Squared measures the proportion of variance in Box Plus/Minus (BPM), explained by the predictors. Essentially, a number to determine how well the model fits the data. This is a value between zero and one, with one being the best estimate of the variance whilst zero is the worst. The model's R-squared value was 0.4588, which indicates that the model explains some of the variability in BPM. In the context of the NBA, player performance is influenced by many different factors, thus, the predictors capture a small portion of the factors contributing to BPM. A caveat with using R-squared is that it either increases or stays the same with additional predictors. As a result, models with many predictors appear to perform better based on this metric alone. The model features seven predictors, and the data set contains a sizable number of observations. This large ratio of observations to predictions reduces the likelihood of overfitting.

Adjusted R-Squared is similar to R squared but with the additional property of punishing excessive predictors which do not improve model performance. The adjusted R-squared value was 0.4532. This again indicates the model's lower explanatory power of BPM's variance. However, the adj. R-squared value is 0.0056 away from R-squared, conveying the fact that the predictors used in the final model provided meaningful information to predict BPM.

Mean Absolute Error (MAE) value was 2.0165. This measures the average magnitude of predicted errors. In the context of the problem, MAE measures the average difference between actual BPM and predicted BPM. The benefit of MAE is the use of absolute values as opposed to squares which reduce the effect of outliers and are thus less prominent. In the NBA, star players may skew other metrics which utilize squared values disproportionately but MAE provides a more unbiased view. A value of 2.0165 indicates the model is moderately accurate since BPM varies substantially.

The Root Mean Squared Error (RMSE) was 2.6661. This metric is like MAE, but it is swayed more intensely by outliers and large deviations in the data. Its value is in terms of the same units of the response variable (BPM). The RMSE is moderately larger than MAE, signifying some larger residuals exist. However, the difference is not extreme, showing that the outliers exist but are not excessive.

Interpretation of the final model coefficients

The final multiple linear regression model was used to examine how common player statistics are associated with Box Plus/Minus (BPM), a metric intended to reflect a player's overall impact on team performance while on the court. In this setting, each coefficient is interpreted while holding the other predictors fixed, so the model isolates the unique association

between each variable and BPM. Overall, the results suggest that BPM is not driven by scoring alone, but instead reflects a combination of scoring, playmaking, rebounding, age, and positional role.

First, points per game (PTS) has a positive estimated coefficient of 0.169. This indicates that, after adjusting for age, position, assists, and offensive rebounds, a one-point increase in scoring per game is associated with an average increase of 0.169 in BPM. This result shows that scoring remains an important part of player impact, since players who contribute more points tend to have higher BPM values. However, the magnitude of this coefficient is smaller than those for steals and blocks, suggesting that defensive contributions play a larger role in determining BPM than scoring alone.

Next, assists per game (AST) has a coefficient of 0.238. This means that, holding the other predictors constant, one additional assist per game is associated with an average increase of about 0.238 in BPM. Relative to PTS, this larger coefficient suggests that playmaking has a stronger relationship with BPM on a per-unit basis. Substantively, this is reasonable because assists capture a player's ability to create efficient scoring opportunities for teammates and improve overall offensive flow. Therefore, the model suggests that BPM places substantial value on offensive creation in addition to direct scoring.

A similar but even stronger pattern appears for steals per game (STL), which has the largest continuous coefficient in the model at 1.865. Interpreted directly, this means that one additional steal per game is associated with an average increase of about 1.865 in BPM, after controlling for the other variables. This reflects the outsized impact of defensive playmaking and turnover generation on team performance. As a result, STL appears to be the strongest continuous predictor of BPM in the final model.

Age has a small but statistically significant positive effect (0.091, $p < 0.001$), suggesting that veteran players tend to have marginally higher BPM when other statistics are held constant. While the effect size is modest relative to the statistical predictors, the significance of the result suggests that experience and maturity contribute something beyond raw production. This could reflect improved decision-making or efficiency that develops over a player's career..

Position also remains relevant in the final model, but its interpretation requires more care. As position is included as a categorical predictor, the corresponding coefficients represent differences in expected BPM relative to the omitted reference position, after adjusting for age, points, assists, and offensive rebounds. Thus, the position terms should not be interpreted as absolute BPM values for each role. Instead, they indicate that positional role continues to matter even after controlling for standard box-score production. This is consistent with basketball logic, since players at different positions are often asked to contribute in different ways. For example, big men may add value through rebounding and interior presence, while guards may contribute more through ball handling and playmaking.

Taken together, we are provided with a specific view, as the results indicate that the player statistics most strongly associated with BPM are not limited to scoring. Rather, steals and blocks show especially strong positive relationships with BPM, reflecting the outsized value of defensive contributions. Assists and offensive rebounds also contribute meaningfully, while points per game contributes positively but to a smaller extent. Therefore, the players who appear to have the greatest on-court impact are not simply the highest scorers, but those who also defend, create opportunities for teammates, and generate extra possessions. In that sense, the model suggests that defensive playmaking, teamwork, and second-chance creation are central components of individual player impact.

Comparison of the final model results with the literature

The final model results are broadly consistent with the literature reviewed in the project. In particular, they support the general conclusion that player impact is best understood through a multivariable framework rather than through any single statistic in isolation. The positive effects of points, steals, blocks, assists, and offensive rebounds show that BPM reflects several dimensions of performance at once, rather than rewarding scoring alone.

First, the results are consistent with Ghimire, Ehrlich, and Sanders (2020), who argue that plus-minus style outcomes should be studied using multiple predictors rather than one summary measure alone. The present findings follow the same pattern. Although scoring remains important, steals and blocks appear to have even stronger associations with BPM, indicating that

player impact depends on a broader mix of offensive and defensive contributions rather than scoring alone. This supports the idea that individual value in basketball is multidimensional.

Second, the findings also align with Gong and Chen (2024), who emphasize that both performance metrics and position influence plus-minus ratings. That conclusion is reflected in the final model, where position continues to matter even after controlling for age and box-score variables. This suggests that the player role still provides useful information beyond raw production. In other words, how a player contributes appears to depend not only on measurable statistics, but also on the position-specific responsibilities attached to that player's role.

Finally, the model is consistent with Sarlis and Tjortjis (2024), who conclude that traditional box-score statistics remain strong predictors of advanced basketball metrics. This is one of the clearest findings in the present analysis. Even though BPM is an advanced measure, it remains strongly associated with common statistics such as steals, blocks, assists, offensive rebounds, and points per game. In particular, the large STL and BLK coefficients suggest that advanced impact measures still rely heavily on observable box-score contributions, especially those tied to possession creation and offensive organization.

At the same time, there are some differences between this project and the studies in the literature. The cited papers use more advanced modeling approaches, including regularized regression and hierarchical methods, whereas this project uses a simpler multiple linear regression framework. As a result, the present analysis is narrower in scope and focuses mainly on direct, interpretable associations between player-level predictors and BPM. It does not explicitly account for lineup context, team structure, or more complex interactions. Nevertheless, this simpler approach remains useful because it produces results that are easier to interpret while still supporting the main conclusions of the literature.

Overall, the final model fits well within the broader basketball analytics literature. Across both this project and the cited studies, the same general conclusion emerges: player impact is shaped by multiple aspects of performance, position remains relevant, and traditional box-score statistics continue to be meaningful predictors of advanced impact measures. Most importantly, the present results reinforce that BPM is associated not only with scoring, but also with defensive playmaking like steals and blocks, alongside offensive creation and second-chance opportunities.

Discussion and conclusion:

In this project, we tried to answer an important question in basketball: which player statistics best explain how a team performs when that player is on the court, using Box Plus/Minus (BPM) as the measure. By using data from the 2025–26 NBA season and applying multiple linear regression, we looked at how common stats like points, assists, rebounds, and age are related to a player's overall impact. Overall, our results show that basic basketball statistics are still very useful, but some matter more than others.

One of the main conclusions from our analysis is that defensive actions have a strong effect on BPM. Steals (STL) had the largest impact in our model, with a coefficient of 1.865. This means that players who generate turnovers through steals tend to have outsized positive impact on their team's performance. This makes sense because steals directly create extra possessions for the offense while simultaneously denying the opponent a scoring opportunity, a two-way contribution that box-score statistics rarely capture as cleanly. Even though steals are sometimes overlooked compared to scoring, our results show they are the single strongest predictor of BPM among the statistics considered.

Assists (AST) also had a strong positive relationship with BPM. This shows that players who create scoring chances for their teammates are very valuable. Assists reflect teamwork and good decision-making, which are key parts of successful basketball. This finding matches what we see in real games, where teams with good ball movement often perform better. It also supports past research that highlights the importance of playmaking.

Points per game (PTS) did have a positive effect on BPM, but its impact was smaller compared to assists and offensive rebounds. This suggests that scoring alone does not fully explain a player's value. A player can score a lot of points, but if they are not helping in other areas, their overall impact may not be as strong. This shows that looking at only scoring can be misleading when evaluating players.

Age had a small but meaningful effect in our model. While the coefficient is modest compared to the statistical predictors, it was statistically significant, suggesting that veteran players tend to have slightly higher BPM when other factors are held constant. This could reflect

the improved decision-making and efficiency that develops over a career, even if the effect is not as dramatic as scoring or defensive actions.

The model diagnostics also help us understand how reliable our results are. The residual plots show that a linear model is a reasonable choice, since there is no strong pattern in the data. However, the scale–location plot suggests that the spread of errors is not constant, which means the model may not be perfect. The Q–Q plot shows that most of the data follows a normal pattern, but there are some extreme values. This means that very high-performing or very low-performing players may affect the results slightly. The leverage plot shows that there are not many highly influential points, but a few should still be checked carefully.

From these results, we can identify a few key findings. First, traditional statistics like assists and rebounds are still very useful for understanding player impact. Even with advanced metrics like BPM, simple stats continue to provide strong information. Second, not all stats are equally important. Team-based contributions, like passing and rebounding, seem to matter more than scoring alone. Third, player performance should be looked at from multiple angles, since no single stat tells the whole story.

These findings have useful real-world applications. For teams and coaches, the results suggest that focusing only on scoring is not enough. Players who contribute in other ways, such as rebounding or passing, can be just as important or even more important. This can help teams make better decisions when choosing players or building lineups. For example, a player who gets offensive rebounds and creates assists may improve team performance more than a player who only scores.

In terms of strategy, teams may want to focus more on ball movement and second-chance opportunities. Coaches could design plays that increase assists or put players in positions to get offensive rebounds. Player development programs can also use this information to help players improve in these areas, making them more valuable overall.

However, there are some limitations in our study. One limitation is that we only used box-score data. While these stats are easy to understand, they do not include everything that happens in a game. For example, they do not capture defense positioning, off-ball movement, or

teamwork that does not show up in the stats. Using more advanced data, such as player tracking, could improve the analysis.

Another limitation is that we used a simple linear regression model. While this method is easy to interpret, it may not fully capture complex relationships between variables. For example, the effect of assists might be different depending on a player's position or role. Future studies could include interaction terms or use more advanced models to better understand these relationships.

We also did not include team context in our model. A player's BPM can be affected by their teammates, coaching, and overall team quality. Players on strong teams may appear better because they are surrounded by good players. Future research could include team-level variables or use more advanced metrics to adjust for this.

There are several ways this analysis could be improved in the future. First, we could include more variables, such as shooting efficiency or defensive stats, to get a more complete picture of performance. Second, we could look at data over multiple seasons to reduce the effect of random variation. Third, we could group players by position or role to see if the relationships change for different types of players. Finally, using more advanced models or machine learning methods could help find deeper patterns in the data.

In conclusion, this project shows that common basketball statistics are still useful for explaining player impact, even when using advanced metrics like BPM. Steals and blocks were found to have the strongest relationships with team performance, while assists and offensive rebounds also contributed positively, and scoring had a smaller effect than expected. These results show the importance of teamwork and possession-based play in basketball. At the same time, the study also highlights the limits of simple models and basic data. Future research can build on this work by using better data and more advanced methods to better understand what makes players truly valuable.

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- Introduction and Discussion and Conclusion

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